

Dr. Arie Wahyu Wijayanto

Associate Professor, Data Science & Artificial Intelligence | Jakarta, Indonesia

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PROFESSIONAL PROFILE

Dr. Arie Wahyu Wijayanto is an Associate Professor of Data Science and Artificial Intelligence at Politeknik Statistika STIS, Indonesia, and Director of AI Talent Development at KORIKA — Indonesia's national AI collaborative research and industrial innovation association. His career spans two decades at the intersection of AI, digital transformation, and public policy, with a sustained focus on applying geospatial big data and machine learning to solve pressing development challenges across Southeast Asia and the region.

He serves as an appointed Expert Committee Member for national AI competency standards (Ministry of Communications and Digital Affairs) and for Indonesia's National Task Force on Digital Talent and AI Development (Coordinating Ministry for Human Development and Cultural Affairs). An award recipient of the GEO SDGs Award 2023 (Special Category for Innovation) for his "Granular Poverty Mapping from Space" initiative, he has partnered with the United Nations, Asian Development Bank, UNICEF, and leading universities across ASEAN to translate data science into impactful regional policy.

EDUCATION

Tokyo Institute of Technology <i>Doctor of Engineering — Computer Science</i>	<i>Tokyo, Japan</i> 2016 – 2020
Institut Teknologi Bandung (ITB) <i>Master of Engineering — Informatics / Computer Science</i>	<i>Bandung, Indonesia</i> 2013 – 2015
Institute of Statistics (STIS) <i>Bachelor of Applied Science — Computational Statistics</i>	<i>Jakarta, Indonesia</i> 2004 – 2008

ACADEMIC & RESEARCH POSITIONS

Politeknik Statistika STIS <i>Associate Professor — Data Science & Artificial Intelligence</i> <i>Vice Director of Academic Affairs</i> <i>Head, Center for Research and Community Services (PPPM)</i>	<i>Jakarta, Indonesia</i> March 2026 – Present June 2026 - Present September 2022 – May 2026
Drives the institution's research agenda and community engagement mission by forging strategic partnerships with universities, government agencies, and research institutions across Indonesia and global collaboration, cultivating an interdisciplinary research excellence.	
University of Indonesia (UI) <i>Visiting Lecturer</i>	<i>Jakarta, Indonesia</i> August 2022 – Present
Delivers Introduction to Data Science (Faculty of Economics and Business), Unstructured Data Analysis, and Softskills for Data Science (Faculty of Mathematics and Natural Sciences).	
Central Bank of Indonesia (Bank Indonesia) <i>Research Scholar & Partner</i>	<i>Jakarta, Indonesia</i> April 2025 – Present
Provides expert research consultation in AI, data science, geospatial big data and satellite imagery analytics to the Bank Indonesia Institute, supporting evidence-based strategic policy formulation.	

POLICY & ADVISORY ROLES

United Nations Regional Hub on Big Data & Data Science for Asia-Pacific <i>Seminar Programme Secretariat</i>	September 2023 – Present
Managed seminars, webinars, and workshops on data science for statisticians, data scientists, and public sector staff across the Asia-Pacific region, in collaboration with UN Statistics Division, UN ESCAP, FAO, UN Statistical Institute for Asia and the Pacific (SIAP), and University of New South Wales.	

KORIKA — Collaborative Research and Industrial Innovation in Artificial Intelligence

Director of AI Talent Development

January 2025 – Present

Director of AI Curriculum

September 2023 – December 2024

- Designs and leads Indonesia's national AI talent development strategy, including education programs, industry partnerships (Asian Development Bank, Google, AVPN Singapore), and in collaboration with Ministry of ICT, Ministry of Education, academic leaders, and industry partners.

Coordinating Ministry for Human Development and Cultural Affairs

Indonesia

Expert Committee Member — National Task Force on Digital Talent & AI Development

June 2025 – Present

- Provides strategic and technical advisory input on AI workforce development, competency frameworks, higher education alignment, and responsible AI governance at national scale.

Ministry of Communications and Digital Affairs, Republic of Indonesia

Expert Committee — National Work Competency Standards (SKKNI) in AI

May 2025 – Present

- Contributes to formulation of Indonesia's national AI competency frameworks and job standards, ensuring alignment with global best practices and supporting digital talent development.

ASIIN e.V. (Germany)

International

Accreditation Expert

March 2026 – Present

- Independently assesses higher education degree programmes against international quality standards for Germany's leading international accreditation agency.

UN Economic Commission for Europe (UNECE) — Machine Learning Group

Europe

Workstream Lead — Satellite Imagery Analysis Theme

February – November 2021

- Co-led the satellite imagery analysis workstream for official statistics, coordinating machine learning experts from national statistical offices across 9 countries (Finland, Netherlands, France, Indonesia, Malaysia, Ireland, Italy, Switzerland, Ukraine).

SELECTED RESEARCH PROJECTS

ASEAN HPC Trial of Fugaku - Poverty Mapping Modelling — RIKEN, Japan & A*STAR Singapore

Principal Investigator

July 2022 – December 2023

- Developed spatially granular poverty indices using multi-source satellite imagery (VIIRS, Sentinel-2, Sentinel-5P, MODIS) and ResNet CNN transfer learning on the Fugaku Supercomputer (Riken, Japan). Funded by A*STAR Computational Resource Centre.

Granular Poverty Mapping — Asian Development Bank / Japan Fund (JFPR)

Main Researcher

July 2022 – December 2023

- Developed granular poverty maps using ML to empower policymakers in delivering improved programs for vulnerable populations. In collaboration with BPS-Statistics Indonesia, funded by ADB and JFPR.

Child Poverty Mapping in Indonesia — UNICEF and Statistics Indonesia (BPS)

Main Researcher

June – December 2021

- Investigated innovative approaches for mapping child poverty statistics using satellite imagery and geospatial big data, in collaboration with BPS-Statistics Indonesia. Funded by UNICEF.

PROFESSIONAL CERTIFICATION & CAPACITY BUILDING

LSP Politeknik Statistika STIS

Chief of Professional Certification

September 2021 – September 2024

- Founded and led the Professional Certification Institute; obtained nationally-accredited license from Indonesia's Professional Certification Authority (BNSP). Constructed nationally-accredited competency certification schemes for Data Scientist and Associate Data Scientist.
- Certified more than 95% of graduates with professional competency certification (national benchmark: 90%); upgraded 20 internal assessors.

United Nations and ASEAN Secretariat — Certified Trainer

- Trainer, UN Workshop on Big Data (multiple editions). UN Statistics Division.
- Trainer, ASEAN Regional Workshop on Big Data for Enhancing Official Statistics: Supervised Learning with Python. ASEAN Secretariat.
- Trainer, UN Workshop on Data Access and Machine Learning

HONOURS & AWARDS

- GEO SDGs Award 2023 — Special Category for Innovation (“Granular Poverty Mapping from Space”) — Group on Earth Observations (NASA, ESA, JAXA, CSIRO, IEEE, WHO, UNITAR)
- Best Paper Award — 2nd International Conference on Data Science and Official Statistics (ICDSOS) 2023
- Best Paper Award — IEEE International Conference on Communication, Networks, and Satellite (COMNETSAT) 2022
- Best Paper & Presentation Award — IEEE CyberneticsCom 2022
- Best Presentation Award — 1st ICDSOS 2021
- Top 4 Winner — AI-Based Road Damage Detection, Ministry of Public Works, Indonesia
- Exemplary Statistical Person (Insan Statistik Teladan) — BPS-Statistics Indonesia 2023
- Best Paper Award — National Seminar of Official Statistics 2020
- Scholarship Awardee (multiple)

SELECTED SCIENTIFIC PUBLICATIONS (Complete List: <https://s.id/Arie-GScholar>)

1. Sakti A.D.; Wijayanto, Arie Wahyu; et al. (2024). Geospatial intelligence framework for BTS infrastructure planning toward universal internet access in Indonesia. *International Journal of Applied Earth Observation and Geoinformation*, 135, 104274. <https://doi.org/10.1016/j.jag.2024.104274>
2. Utami, N.A.; Wijayanto, Arie Wahyu; Pramana, S.; Astuti, E.T. (2023). Spatially granular poverty index (SGPI) for urban poverty mapping in Jakarta metropolitan area: a remote sensing satellite imageries and geospatial big data approach. *Earth Science Informatics*, 16, 3531–3544. Springer. <https://doi.org/10.1007/s12145-023-01084-7>
3. Putri, S.R.; Wijayanto, Arie Wahyu; Pramana, S. (2023). Multi-source satellite imagery and point of interest data for poverty mapping in East Java: Machine learning and deep learning approaches. *Remote Sensing Applications: Society and Environment*, 29, 100889. Elsevier. <https://doi.org/10.1016/j.rsase.2022.100889>
4. Putra, Y.C.; Wijayanto, Arie Wahyu. (2023). Automatic detection and counting of oil palm trees using remote sensing and object-based deep learning. *Remote Sensing Applications: Society and Environment*, 29, 100914. Elsevier. <https://doi.org/10.1016/j.rsase.2022.100914>
5. Nurkarim, W.; Wijayanto, Arie Wahyu. (2022). Building footprint extraction and counting on very high-resolution satellite imagery using object detection deep learning. *Earth Science Informatics*. Springer. <https://doi.org/10.1007/s12145-022-00895-4>
6. Putra, Y.C.; Wijayanto, Arie Wahyu; Chulafak, G.A. (2022). Oil palm trees detection and counting using VHR satellite imagery and UAV data. *Ecological Informatics*. Elsevier. <https://doi.org/10.1016/j.ecoinf.2022.101878>
7. Kurniawan, R.; Wijayanto, Arie Wahyu; et al. (2022). Eco-environment vulnerability assessment using remote sensing in East Kalimantan, Indonesia. *Remote Sensing Applications: Society and Environment*, 27, 100791. Elsevier. <https://doi.org/10.1016/j.rsase.2022.100791>
8. Daulay, N.A.; Wijayanto, Arie Wahyu; et al. (2024). Optimizing Malaria Control: Granular and Cost-Effective Mosquito Habitat Index in Endemic Areas. *Knowledge Engineering and Data Science*, 7(1), 40–57.
9. Kadarisman, M.; Wijayanto, Arie Wahyu; Sakti, A.D. (2022). Government Agencies' Readiness Evaluation towards Industry 4.0 and Society 5.0 in Indonesia. *Social Sciences*, 11(8), 331. MDPI.
10. Ramadhan, R.; Wijayanto, Arie Wahyu. (2023). Integrating Satellite Imageries and Multiple Geospatial Big Data for Granular Mapping of Human Development Index in East Java. *Proceedings of ICDSOS*.

LANGUAGES

Indonesian
Japanese

Native
Limited Working Proficiency

English Professional Working Proficiency